

ITA 0502- COMPUTER VISION

LAB EXPERIMENT-13

13. Perform Perspective Transformation on the Video.

CODING:

```
import cv2
import numpy as np

# Open the video
cap = cv2.VideoCapture("video.mp4")

while True:

    ret, frame = cap.read()

    if not ret:
        break

    h, w = frame.shape[:2]
    # Source points
    pts1 = np.float32([
        [50, 50],
        [w-50, 50],
        [50, h-50],
        [w-50, h-50]
    ])

    # Destination points
    pts2 = np.float32([
        [0, 100],
        [w, 0],
        [100, h],
        [w-100, h-100]
```

)

```
# Perspective matrix
```

```
M = cv2.getPerspectiveTransform(pts1, pts2)
```

```
# Apply transformation
```

```
result = cv2.warpPerspective(frame, M, (w, h))
```

```
# Show original and transformed video
```

```
cv2.imshow("Original Video", frame)
```

```
cv2.imshow("Perspective Video", result)
```

```
# Press q to exit
```

```
if cv2.waitKey(30) & 0xFF == ord('q'):
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

OUTPUT:

